

Node.js Module System – Complete Notes

1. What is a module in Node.js?

Every JavaScript file in Node.js is treated as a module.
Each file has its own scope and can export values.

2. Core Concept

Node.js does NOT execute a JS file directly.
It wraps every file inside a function and then executes it.
This is called the Module Wrapper Function.

3. Internal Wrapper

```
(function (exports, require, module, __filename, __dirname) {  
  // code from xyz.js  
});
```

4. Why Wrapper is Used

- Provides require()
- Provides module and exports
- Gives __filename and __dirname
- Ensures file-level scope isolation

5. Text Flow Diagram

You write a JavaScript file

```
xyz.js  
const x = 10;  
module.exports = x;
```

Node.js Runtime

wraps the file inside a function and executes it

Module Execution

Variables are private
require() is available
module.exports is available

Importing

```
const x = require('./xyz');
```

6. Why variables are not global

Because the file runs inside a function, variables remain private.

7. require() working

Loads file → wraps → executes once → returns exports → caches result

8. Interview Summary

Node.js treats every file as a module by wrapping it in a function,
giving private scope and module features.